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{%-% macro format_parameters(properties, required, filter_keys=false) -%}
  {%-% set standard_keys = ['description', 'type', 'properties', 'required', 'nullable'] -%}
  {%-% set ns = namespace(found_first=false) -%}
  {%-% for key, value in properties | dictsort -%}
    {%-% set add_comma = false -%}
    {%-% if not filter_keys or key not in standard_keys -%}
      {%-% if ns.found_first %, {%-% endif -%}
      {%-% set ns.found_first = true -%}
      {{ key }}:{
        {%-% if value['description'] -%}
          description:<|">{{ value['description'] }}<|">
          {%-% set add_comma = true -%}
        {%-% endif -%}
        {%-% if value['type'] | upper == 'STRING' -%}
          {%-% if value['enum'] -%}
            {%-% if add_comma %, {%-% else -%} {%-% set add_comma = true -%} {%-% endif -%}
            enum:{{ format_argument(value['enum']) }}
          {%-% endif -%}
        {%-% elif value['type'] | upper == 'ARRAY' -%}
          {%-% if value['items'] is mapping and value['items'] -%}
            {%-% if add_comma %, {%-% else -%} {%-% set add_comma = true -%} {%-% endif -%}
            items:{
              {%-% set ns_items = namespace(found_first=false) -%}
              {%-% for item_key, item_value in value['items'] | dictsort -%}
                {%-% if item_value is not none -%}
                  {%-% if ns_items.found_first %, {%-% endif -%}
                  {%-% set ns_items.found_first = true -%}
                  {%-% if item_key == 'properties' -%}
                    properties:{
                      {%-% if item_value is mapping -%}
                        {{- format_parameters(item_value, value['items']['required'] | default([])) -}}
                      {%-% endif -%}
                    }
                  {%-% elif item_key == 'required' -%}
                    required:[
                      {%-% for req_item in item_value -%}
                        <|">{{- req_item -}}<|">
                        {%-% if not loop.last %, {%-% endif -%}
                      {%-% endfor -%}
                    ]
                  {%-% elif item_key == 'type' -%}
                    {%-% if item_value is string -%}
                      type:{{ format_argument(item_value | upper) }}

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        {%- else -%}
        type:{{ format_argument(item_value | map('upper') | list) }}
        {%- endif -%}
    {%- else -%}
    {{ item_key }}:{{ format_argument(item_value) }}
    {%- endif -%}
{%- endif -%}
{%- endfor -%}
}
{%- endif -%}
{%- endif -%}
{%- if value['nullable'] %}
    {%- if add_comma %},{%- else -%} {%- set add_comma = true -%} {%- endif -%}
    nullable:true
{%- endif -%}
{%- if value['type'] | upper == 'OBJECT' -%}
    {%- if value['properties'] is defined and value['properties'] is mapping -%}
        {%- if add_comma %},{%- else -%} {%- set add_comma = true -%} {%- endif -%}
        properties:{
            {{- format_parameters(value['properties'], value['required'] | default([])) -}}
        }
    {%- elif value is mapping -%}
        {%- if add_comma %},{%- else -%} {%- set add_comma = true -%} {%- endif -%}
        properties:{
            {{- format_parameters(value, value['required'] | default([]), filter_keys=true) -}}
        }
    {%- endif -%}
    {%- if value['required'] -%}
        {%- if add_comma %},{%- else -%} {%- set add_comma = true -%} {%- endif -%}
        required:[
            {%- for item in value['required'] | default([]) -%}
                <|">{{- item -}}<|">
            {%- if not loop.last %},{%- endif -%}
            {%- endfor -%}
        ]
    {%- endif -%}
{%- endif -%}
{%- if add_comma %},{%- else -%} {%- set add_comma = true -%} {%- endif -%}
type:<|">{{ value['type'] | upper }}<|">
{%- endif -%}
{%- endfor -%}
{%- endmacro -%}
{%- macro format_function_declaration(tool_data) -%}

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    declaration:{{- tool_data['function']['name'] -}}{description:<|">{{-
tool_data['function']['description'] -}}<|">
    {%- set params = tool_data['function']['parameters'] -%}
    {%- if params -%}
        ,parameters:{
            {%- if params['properties'] -%}
                properties:{{- format_parameters(params['properties'], params['required']) -}} },
            {%- endif -%}
            {%- if params['required'] -%}
                required:[
                    {%- for item in params['required'] -%}
                        <|">{{- item -}}<|">
                        {{- ', ' if not loop.last -}}
                    {%- endfor -%}
                ],
            {%- endif -%}
            {%- if params['type'] -%}
                type:<|">{{- params['type'] | upper -}}<|">
            {%- endif -%}
        {%- endif -%}
    {%- if 'response' in tool_data['function'] -%}
        {%- set response_declaration = tool_data['function']['response'] -%}
        ,response:{
            {%- if response_declaration['description'] -%}
                description:<|">{{- response_declaration['description'] -}}<|">,
            {%- endif -%}
            {%- if response_declaration['type'] | upper == 'OBJECT' -%}
                type:<|">{{- response_declaration['type'] | upper -}}<|">
            {%- endif -%}
        {%- endif -%}
    }
{%- endmacro -%}
{%- macro format_argument(argument, escape_keys=True) -%}
    {%- if argument is string -%}
        {{- '<|">' + argument + '<|">' -}}
    {%- elif argument is boolean -%}
        {{- 'true' if argument else 'false' -}}
    {%- elif argument is mapping -%}
        {{- '{' -}}
        {%- set ns = namespace(found_first=false) -%}
        {%- for key, value in argument | dictsort -%}
            {%- if ns.found_first %},{% endif -%}
            {%- set ns.found_first = true -%}

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    {%- if escape_keys -%}
        {{- '<|>' + key + '<|>' -}}
    {%- else -%}
        {{- key -}}
    {%- endif -%}
    :{{- format_argument(value, escape_keys=escape_keys) -}}
{%- endfor -%}
{{- '}' -}}
{%- elif argument is sequence -%}
    {{- '[' -}}
    {%- for item in argument -%}
        {{- format_argument(item, escape_keys=escape_keys) -}}
        {%- if not loop.last %},{% endif -%}
    {%- endfor -%}
    {{- ']' -}}
{%- else -%}
    {{- argument -}}
{%- endif -%}
{%- endmacro -%}

{%- macro strip_thinking(text) -%}
    {%- set ns = namespace(result='') -%}
    {%- for part in text.split('<channel>') -%}
        {%- if '<channel>' in part -%}
            {%- set ns.result = ns.result + part.split('<channel>')[0] -%}
        {%- else -%}
            {%- set ns.result = ns.result + part -%}
        {%- endif -%}
    {%- endfor -%}
    {{- ns.result | trim -}}
{%- endmacro -%}

{%- macro format_tool_response_block(tool_name, response) -%}
    {{- '<|tool_response>' -}}
    {%- if response is mapping -%}
        {{- 'response:' + tool_name + '{' -}}
        {%- for key, value in response | dictsort -%}
            {{- key -}}:{{- format_argument(value, escape_keys=False) -}}
            {%- if not loop.last %},{% endif -%}
        {%- endfor -%}
        {{- '}' -}}
    {%- else -%}
        {{- 'response:' + tool_name + '{value:' + format_argument(response, escape_keys=False) +
    '}' -}}

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    {%- endif -%}
    {{- '<tool_response|>' -}}
{%- endmacro -%}

{%- set ns = namespace(prev_message_type=None) -%}
{%- set loop_messages = messages -%}
{{- bos_token -}}
{%- Handle System/Tool Definitions Block -#}
{%- if (enable_thinking is defined and enable_thinking) or tools or messages[0]['role'] in
['system', 'developer'] -%}
    {{- '<|turn>system\n' -}}
    {%- Inject Thinking token at the very top of the FIRST system turn -#}
    {%- if enable_thinking is defined and enable_thinking -%}
        {{- '<|think|>\n' -}}
        {%- set ns.prev_message_type = 'think' -%}
    {%- endif -%}
    {%- if messages[0]['role'] in ['system', 'developer'] -%}
        {%- if messages[0]['content'] is string -%}
            {{- messages[0]['content'] | trim -}}
        {%- elif messages[0]['content'] is sequence -%}
            {%- for item in messages[0]['content'] -%}
                {{- item['text'] | trim + ' ' -}}
            {%- endfor -%}
        {%- endif -%}
        {%- set loop_messages = messages[1:] -%}
    {%- endif -%}
    {%- if tools -%}
        {%- for tool in tools %}
            {{- '<|tool>' -}}
            {{- format_function_declaration(tool) | trim -}}
            {{- '<tool|>' -}}
        {%- endfor %}
        {%- set ns.prev_message_type = 'tool' -%}
    {%- endif -%}
    {{- '<turn|>\n' -}}
{%- endif %}

{%- Pre-scan: find last user message index for reasoning guard -#}
{%- set ns_turn = namespace(last_user_idx=-1) -%}
{%- for i in range(loop_messages | length) -%}
    {%- if loop_messages[i]['role'] == 'user' -%}
        {%- set ns_turn.last_user_idx = i -%}
    {%- endif -%}

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{%-% endfor -%}

{%- Loop through messages -#}
{%-% for message in loop_messages -%}
    {%-% if message['role'] != 'tool' -%}
    {%-% set ns.prev_message_type = None -%}
    {%-% set role = 'model' if message['role'] == 'assistant' else message['role'] -%}
    {%- Detect continuation: suppress duplicate <|turn>model when previous non-tool message
was also assistant -#}
    {%-% set prev_nt = namespace(role=None, found=false) -%}
    {%-% if loop.index0 > 0 -%}
        {%-% for j in range(loop.index0 - 1, -1, -1) -%}
            {%-% if not prev_nt.found -%}
                {%-% if loop_messages[j]['role'] != 'tool' -%}
                    {%-% set prev_nt.role = loop_messages[j]['role'] -%}
                    {%-% set prev_nt.found = true -%}
                {%-% endif -%}
            {%-% endif -%}
        {%-% endfor -%}
    {%-% endif -%}
    {%-% set continue_same_model_turn = (role == 'model' and prev_nt.role == 'assistant') -%}
    {%-% if not continue_same_model_turn -%}
        {{- '<|turn>' + role + '\n' }}
    {%-% endif -%}

    {%- Render reasoning/reasoning_content as thinking channel -#}
    {%-% set thinking_text = message.get('reasoning') or message.get('reasoning_content') -%}
    {%-% if thinking_text and loop.index0 > ns_turn.last_user_idx and message.get('tool_calls') -%}
        {{- '<|channel>thought\n' + thinking_text + '\n<channel|>' -}}
    {%-% endif -%}

    {%-% if message['tool_calls'] -%}
        {%-% for tool_call in message['tool_calls'] -%}
            {%-% set function = tool_call['function'] -%}
            {{- '<|tool_call>call:' + function['name'] + '{' -}}
            {%-% if function['arguments'] is mapping -%}
                {%-% set ns_args = namespace(found_first=false) -%}
                {%-% for key, value in function['arguments'] | dictsort -%}
                    {%-% if ns_args.found_first %}{%-% endif -%}
                    {%-% set ns_args.found_first = true -%}
                    {{- key -}}:{{- format_argument(value, escape_keys=False) -}}
                {%-% endfor -%}
            {%-% elif function['arguments'] is string -%}

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        {{- function['arguments'] -}}
    {%- endif -%}
    {{- '<tool_call|>' -}}
{%- endfor -%}
{%- set ns.prev_message_type = 'tool_call' -%}
{%- endif -%}

{%- set ns_tr_out = namespace(flag=false) -%}
{%- if message.get('tool_responses') -%}
    {%- Legacy: tool_responses embedded on the assistant message (Google/Gemma
native) -#}
    {%- for tool_response in message['tool_responses'] -%}
        {{- format_tool_response_block(tool_response['name'] | default('unknown'),
tool_response['response']) -}}
        {%- set ns_tr_out.flag = true -%}
        {%- set ns.prev_message_type = 'tool_response' -%}
    {%- endfor -%}
{%- elif message.get('tool_calls') -%}
    {%- OpenAI Chat Completions: forward-scan consecutive role:tool messages -#}
    {%- set ns_tool_scan = namespace(stopped=false) -%}
    {%- for k in range(loop.index0 + 1, loop_messages | length) -%}
        {%- if ns_tool_scan.stopped -%}
            {%- elif loop_messages[k]['role'] != 'tool' -%}
                {%- set ns_tool_scan.stopped = true -%}
            {%- else -%}
                {%- set follow = loop_messages[k] -%}
                {%- Resolve tool_call_id to function name -#}
                {%- set ns_tname = namespace(name=follow.get('name') | default('unknown'))
-%}

                {%- for tc in message['tool_calls'] -%}
                    {%- if tc.get('id') == follow.get('tool_call_id') -%}
                        {%- set ns_tname.name = tc['function']['name'] -%}
                    {%- endif -%}
                {%- endfor -%}
                {%- Handle content as string or content-parts array -#}
                {%- set tool_body = follow.get('content') -%}
                {%- if tool_body is string -%}
                    {{- format_tool_response_block(ns_tname.name, tool_body) -}}
                {%- elif tool_body is sequence and tool_body is not string -%}
                    {%- set ns_txt = namespace(s="") -%}
                    {%- for part in tool_body -%}
                        {%- if part.get('type') == 'text' -%}
                            {%- set ns_txt.s = ns_txt.s + (part.get('text') | default("")) -%}
                        {%- else -%}
                            {{- format_tool_response_block(ns_tname.name, part.get('text') | default("")) -}}
                        {%- endfor -%}
                    {%- endfor -%}
                {%- endfor -%}
            {%- endfor -%}
        {%- endfor -%}
    {%- endfor -%}

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        {%- endif -%}
    {%- endfor -%}
    {{- format_tool_response_block(ns_tname.name, ns_txt.s) -}}
    {%- for part in tool_body -%}
        {%- if part.get('type') == 'image' -%}
            {{- '<|image|>' -}}
        {%- elif part.get('type') == 'audio' -%}
            {{- '<|audio|>' -}}
        {%- elif part.get('type') == 'video' -%}
            {{- '<|video|>' -}}
        {%- endif -%}
    {%- endfor -%}
    {%- else -%}
        {{- format_tool_response_block(ns_tname.name, tool_body) -}}
    {%- endif -%}
    {%- set ns_tr_out.flag = true -%}
    {%- set ns.prev_message_type = 'tool_response' -%}
{%- endif -%}
{%- endfor -%}
{%- endif -%}

{%- set captured_content -%}
{%- if message['content'] is string -%}
    {%- if role == 'model' -%}
        {{- strip_thinking(message['content']) -}}
    {%- else -%}
        {{- message['content'] | trim -}}
    {%- endif -%}
{%- elif message['content'] is sequence -%}
    {%- for item in message['content'] -%}
        {%- if item['type'] == 'text' -%}
            {%- if role == 'model' -%}
                {{- strip_thinking(item['text']) -}}
            {%- else -%}
                {{- item['text'] | trim -}}
            {%- endif -%}
        {%- elif item['type'] == 'image' -%}
            {{- '<|image|>' -}}
            {%- set ns.prev_message_type = 'image' -%}
        {%- elif item['type'] == 'audio' -%}
            {{- '<|audio|>' -}}
            {%- set ns.prev_message_type = 'audio' -%}
        {%- elif item['type'] == 'video' -%}

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        {{- '<|video|>' -}}
        {%- set ns.prev_message_type = 'video' -%}
    {%- endif -%}
{%- endfor -%}
{%- endif -%}
{%- endset -%}

{{- captured_content -}}
{%- set has_content = captured_content | trim | length > 0 -%}

{%- if ns.prev_message_type == 'tool_call' and not ns_tr_out.flag -%}
    {{- '<|tool_response>' -}}
    {%- elif not (ns_tr_out.flag and not has_content) -%}
        {{- '<turn|>\n' -}}
    {%- endif -%}
{%- endif -%}
{%- endfor -%}

{%- if add_generation_prompt -%}
    {%- if ns.prev_message_type != 'tool_response' and ns.prev_message_type != 'tool_call' -%}
        {{- '<|turn>model\n' -}}
    {%- endif -%}
{%- endif -%}

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